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Market splitting in Germany – New evidence from a three-stage numerical model of Europe

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HIGHLIGHTS

- A three-stage approach to model redispatch and zonal pricing is proposed.
- Energy transition will lead to a tripling of congestion in Germany until 2020.
- Market splitting in Germany substantially reduces redispatch measures.
- Overall welfare gain is negligible, but considerable distributional effects occur.
- The beneficial effects of market splitting depend strongly on its design.

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ABSTRACT

With the sharp increase in redispatch measures seen during recent years, the importance of an efficient congestion management has increased, particularly in Germany. As the current market design, with a single bidding area, ignores the physical constraints of the transmission grid, there is an ongoing discussion about introducing price zones. Against this background, we develop a three-stage approach to model redispatch and market splitting for Germany while considering interactions with interconnected countries. We identify an increasing spatial imbalance between generation and load and delays in grid extension as being the main drivers for the increase of modeled redispatch volumes from 2012 to 2020. We show that market splitting reduces imminent congestion although results are sensitive to the zonal delimitation and corresponding net transfer capacities. The overall welfare effect is negligible, but price differences between the bidding areas investigated, i.e. one Northern and Southern price zone, result in considerable distributional effects. While consumers in Northern Germany would benefit – producer rents and in particular the value of wind energy would decrease – the opposite is true for Southern Germany. We conclude that market splitting constitutes a solution to reduce redispatch measures as long as transmission grid expansion is further delayed.

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1. Introduction

Germany's accelerated nuclear phase out, increasing intermittent electricity production from renewable energy sources (RES) and a grid not keeping pace with these developments, as well as increasing international trading activities, are challenging the European electricity transmission grids and cause increasing difficulties for transmission system operators (TSOs) in daily grid operation. While redispatch[1] in Germany was exceptional in the past, curative congestion management, and especially redispatch, are now more and more necessary to secure grid stability. Since 2011 the number of redispatch measures has substantially increased, albeit from a low level of 1589 redispatch events in 2010.[2] In 2013 the single price zone induced technically infeasible market results of 7965 h[3] forcing the four German TSOs to adjust the market-based generation. The corresponding total redispatch volume covered 4390 GWh while the total redispatch costs amount to 132.6 M€ (cf. BNetzA and BKartA, 2014).

To counter this, provision is made for network expansion as suggested by the German grid development plan (cf. 50Hertz

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[1] ENTSO-E (2012b, p. 9) defines redispatch as follows: "Redispatching means a measure activated by one or several System Operators by altering the generation and/or load pattern, in order to change physical flows in the Transmission System and relieve a Physical Congestion".

[2] The main reason for the increase seen in 2011 was the moratorium on nuclear power and the corresponding shut-down of Germany's eight oldest nuclear power plants.

[3] Refers to redispatch event hours, not to the total number of hours of a year (cf. Section 2.5 for the definition applied).

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Transmission GmbH, 2014b) which is based on the legal requirements as stipulated by the German Energy Management Act. Besides grid investments the allocation of the use of scarce transmission capacity provides an alternative solution to manage congestion. Several congestion management schemes such as nodal, zonal and uniform pricing, combined with redispatch but also market coupling, are currently being discussed in academic literature and policy making (cf. for instance Egerer et al., 2015; Kunz, 2013; Neuhoff et al., 2013; Oggioni and Smeers, 2013).

According to textbook economic theory, the first-best answer for an efficient congestion management is nodal pricing as nodal prices not only reflect marginal generation costs but also take the costs of grid constraints into account (cf. Green, 2007; Hogan, 1992; Schweppe et al., 1988). Harvey and Hogan (2000) underline the benefits of nodal pricing, compared to zonal market designs, with particular regard to competition and local market power. While a nodal representation provides transparent market results, zonal pricing systems include (1) a hypothetical intra-zonal unconstrained dispatch step and (2) a redispatch step (in order to deal with technical infeasibilities arising from the previous intra-zonal unconstrained dispatch) and thus require more complex market rules. Consequently, in these steps market power could arise that would not exist in a nodal market. That is what happened in California from 2000 where new resources entered the market and intra-zonal congestion became very frequent and significant. Strategic bidding behavior, through the withholding of capacity in the dispatch market, together with inefficient forward and redispatch markets, lead to a high increase of electricity prices and congestion costs and finally to a blackout (cf. e.g. Alaywan et al., 2004; Joskow and Kahn, 2002).

However, a timely implementation of nodal pricing in Germany or Europe seems unrealistic for several reasons – notably due to the need for a German or Europe-wide Independent System Operator (ISO) and corresponding political obstacles. The implementation of additional bidding zones with regard to physical transmission constraints, or in other words zonal pricing, as for example implemented in the Scandinavian market Nordpool (cf. Bjørndal and Jörnsten, 2001; 2007), could be a preferable interim solution to deal with increasing congestion in Germany (cf. Breuer et al., 2013). In this case no ISO is needed and a faster implementation of the mechanism can be expected. On a national and European level the topic of an alternative bidding zone configuration is currently intensively discussed and investigated (cf. e.g. BMWi, 2014; ACER, 2014; ENTSO-E, 2014a; Breuer et al., 2013). The draft regulatory framework foresees a periodical review of national bidding zones (cf. European Commission, 2015). In this regard an ENTSO-E initiative is currently examining the adequacy of the present bidding zone configuration in Europe (cf. ENTSO-E, 2014a).

In order to be able to evaluate different congestion management schemes for Germany, the development of the congestion situation and the corresponding redispatch volumes in Germany first have to be adequately modeled. This involves two main challenges. First, resulting load flows and congestion as well as available generation capacity for redispatch are particularly affected by the modeling of operating states (on-/off-mode) of generation units. Hence, the modeling of unit-wise redispatch requires an hourly mixed-integer unit commitment model. Second, the stand-alone modeling of Germany is insufficient as intermittent RES generation, changes in market design (such as the implementation of zonal pricing in Germany), and grid expansion, substantially influence electricity flows within the entire European transmission grid. To deal with these conflicting requirements, we develop a three-stage modeling approach for evaluating congestion and redispatch in Germany while considering interactions with interconnected countries. In a first step, a load flow model is used to compute power flow sensitivities and transfer capacities. Subsequently, the dispatch and resulting cross-border flows for the European power system are modeled using a large scale linear dispatch model. Finally, we use a mixed integer dispatch model for the detailed modeling of generation constraints and to compute congestion and redispatch measures.

The focus of our case study is on Germany in 2020. This time horizon is chosen due to its critical nature. While urgent network measures are scheduled for completion by 2020 according to the national grid development plan (50Hertz Transmission GmbH, 2014b), grid expansion is currently delayed by about three years (cf. BNetzA, 2012). Moreover, the institutional implementation of additional bidding zones in Germany is expected to take about three years (cf. Consentec and Frontier Economics, 2011). Besides the evaluation of the congestion situation from a system operation and an economic point of view, we analyze potential benefits from market splitting and highlight the main issues posing challenges to the successful design and implementation of zonal pricing in Germany. Furthermore, distributional effects of market splitting and implications for the integration of RES are discussed.

The paper is organized as follows. After a short review of the relevant literature focusing on the modeling of redispatch and congestion management, in Section 2 we describe our model framework including the applied load flow and dispatch models. Furthermore, we discuss measures chosen to analyze the impact of market splitting and describe the analyzed scenarios. Our model results are shown in Section 3 where we evaluate changes in the congestion situation in Germany from 2012 to 2020 and assess the impacts of market splitting. Section 4 draws the main conclusions for policy makers.

2. Methodology

2.1. Relevant literature

Future congestion management is one of the major market design issues in the European electricity market. The network code and framework guidelines on capacity allocation and congestion management for electricity (CACM) as proposed by ENTSO-E (2012b) and adopted by the European Commission (2015) are a major step in paving the way for efficient congestion management for the whole of Europe. Due to the increased spatial imbalance between generation and load in combination with insufficient grid expansion, congestion management in Germany is of particular interest. In the future, regional imbalances will continue to increase as wind capacity in Northern Germany is added and the accelerated nuclear phase out combined with imminent shutdowns of gas fired plants being out of the money increases scarcity in Southern Germany. So far it has not been possible to investigate a full European power system with a high (hourly) temporal resolution – due to the high dimensionality of the resulting optimization problem (cf. Breuer et al., 2013). Hence, the modeling of redispatch volumes and costs involves necessary but appropriate simplifications.

For instance, Burstedde (2012) uses the cost-minimizing European linear investment and DC grid model NEULING to quantify the difference in total system costs between a first-best nodal and a zonal electricity market design for Europe. Redispatch costs are also calculated within this context. The high temporal resolution of 8760 h a year and the geographical representation of the core European model regions seem to be adequate simplifications with regard to the European focus of the study. However, the linear programming approach allows no detailed modeling of unit-specific constraints such as on-/off-status, minimum run times or minimum generation. Furthermore, the applied partial

intertemporal optimization of redispatch leads to an overestimation of ramping costs.

Focussing on Germany, Kunz and Zerrahn (2013) analyze the impacts of different degrees of coordination between TSOs on redispatch costs and volumes within a generalized Nash game. Although the hourly resolution of the model, the nodal representation of the German transmission network and the unit-wise dispatch optimization are suitable for the modeling of redispatch, the authors do not take into account constraints resulting from unit commitment restrictions such as ramping constraints, minimum generation and online/offline times. In addition the neglect of intertemporal decisions (e.g. the dispatch of pumped-storage plants not being modeled endogenously) leads to a lower level of flexibility within the system. Furthermore, interactions with interconnected countries are not considered as imports, and exports are modeled exogenously.

Further recent studies have examined the impacts of alternative bidding zones in Germany and Europe. For instance Breuer et al. (2012) analyze the impacts of alternative bidding zones on the Austrian transmission grid and critical transmission lines in Germany, Czech Republic and Poland for 2010. They find that a splitting of the joint German/Austrian market area would not, among other things, lead to a significant reduction in redispatch costs. Egerer et al. (2015) model the impacts of two price zones in Germany on the German power market and analyze distributional effects. They conclude that market splitting leads to a decrease in curative congestion management measures and find that there is a redistribution between consumers and producers of several hundred million Euros per year. However, cross-border effects of two price zones on the European power market are not modeled endogenously, which may distort the distributional effects. Based on a qualitative analysis for 2011, Consentec and Frontier Economics (2011) state that the economic impacts of a potential splitting of the German power market into two bidding zones are limited. Furthermore, Thema Consulting Group and E-Bridge (2012) analyze the impact of internal congestion within Germany and Great Britain on the value of new interconnectors with Norway. Thereby, they assume an establishment of three bidding zones in Germany and two market zones in Great Britain in 2018. They also find the effects of market splitting in their calculations to be limited as there is only low internal congestion in Germany. However, in the contributions mentioned it is assumed that grid extensions according to the German electricity grid expansion act (EnLAG) are completed as planned. Thus, current delays in grid extension of about 3 years are not taken into account (cf. BNetzA, 2012). Moreover, the internal congestion in Germany significantly depends on the expansion and geographical distribution of RES. Yet, feed-in from offshore wind is, apart from Egerer et al. (2015), not included in existing studies with a time frame aligned to 2018, resulting in much lower internal congestion in Germany.

While there is a consensus on the need for further development of the German congestion management system, different approaches are proposed in existing studies encompassing qualitative and quantitative methods – the latter featuring solely linear models. In addition a rolling planning approach aiming to enhance intertemporal modeling has not been applied so far. However, as already mentioned above, the modeling of unit-wise redispatch requires an hourly mixed-integer unit commitment model. This is because resulting load flows and congestion, as well as available generation capacity for redispatch measures, are particularly affected by the modeling of operating states (on-/off-mode) and intertemporal constraints (e.g. ramping and start-up costs) of generation units.

So far there has been no study using such a detailed dispatch optimization model with hourly resolution and rolling planning for Germany given intertemporal unit commitment restrictions as well as interactions with interconnected countries. Therefore, in this study we fill the gap and analyze the extent to which market splitting can potentially contribute to congestion management and the integration of intermittent RES.

2.2. Basic model framework

As indicated above, modeling redispatch for Germany, and assessing the impact of market splitting, require a detailed modeling of the power plant dispatch and the resulting electricity flows with a high (hourly) temporal resolution. Varying production of RES and changes in the German electricity market design, such as the introduction of market splitting, will influence the entire European power system in terms of national electricity prices and resulting imports and exports. This in turn will induce indirect impacts on redispatch. Thus, the stand-alone modeling of Germany with fixed imports and exports is inappropriate in the context of modeling redispatch in Germany. Hence, an intertemporal Europe-wide, nodal, mixed-integer unit commitment model with hourly resolution, using the rolling planning approach would be the best way to meet all these requirements. However, such a complex model would be difficult to handle due to high computation times and, to our knowledge, hasn't so far been compiled.

We therefore develop a three-stage modeling approach that optimizes redispatch for Germany, within acceptable computation times, whilst adequately considering the most important drivers of redispatch measures. The applied load flow model and the optimization approach applied in both variants of the scheduling[4] model WILMAR Joint Market Model (JMM) are described in more detail in the following Sections 2.3 and 2.4.

The basic idea is to combine two model variants of the dispatch model WILMAR JMM: a basic linear model variant for Europe (LP JMM Europe) and a more detailed one for Germany (MIP JMM DE). In a preceding step, net transfer capacities (NTCs)[5], interface capacities (ICs)[6] and power transfer distribution factors (PTDFs)[7] for Germany are calculated using a *load flow model (0) (cf.* Fig. 1*)*. Then the power plant commitment and dispatch is optimized in *LP JMM Europe (1)* under simplified assumptions for most European countries, using NTC values to limit transactions between bidding zones and IC values to limit transmission on the interconnections between (aggregated) nodes. By using a European PTDF matrix, the modeled European trading flows are subsequently transformed into physical import and export flows and used as input factors for the more detailed *MIP JMM DE (2)*. The latter model focuses on the German transmission grid using NTCs, ICs and PTDFs, all calculated in the load flow model.

Small-scale power plants not directly connected to the transmission grid are a particular issue for the modeling. These represent more than one third of the total generation in Germany, nevertheless they are not typically (so far) used for redispatch measures. Therefore, small-scale power plants are not modeled endogenously in MIP JMM DE, yet they are in LP JMM Europe. Consequently, the electricity produced in subordinated grids, as modeled in LP JMM Europe, is used to calculate the load at the

[4] Note that we use the term *scheduling* to designate both *unit commitment* and *dispatch* decisions for power plants. The linear programming model will not provide exact unit commitment decisions, but rather a continuous approximation labeled *capacity online* (see Tuohy et al., 2009 or Weber et al., 2009 for more details).

[5] ENTSO-E (2011) defines the NTC as the maximum commercial exchange program between two interconnected bidding zones, which is possible without compromising system security and taking into account uncertainties on future network conditions. For a detailed description of the determination of NTC values see Bucksteeg et al. (2015).

[6] ICs are derived under N-1 secure operation. We labeled the transmission capacities as interface capacities because we consider flows between aggregated nodes.

[7] By using PTDFs, financial transactions between bidding zones (which result from the day-ahead optimization of the power plant dispatch) can be translated into physical load flows.

transmission grid level (the so-called 'vertical load') which is used in turn as input in MIP JMM DE.

The model LP JMM Europe hence ensures that the main impacts of changes within Germany (RES, conventional power plants, grid extension, market design etc.) are considered within the meshed European electricity network. The model variant MIP JMM DE complements this by modeling unit commitment and dispatch for Germany in detail together with the electricity flows, the resulting congestion and the corresponding redispatch measures.

The main benefits of our integrated approach to model redispatch and assess potential benefits from market splitting can be summarized as follows:

- Interactions between national power systems in Europe caused by changes in market design such as market splitting are considered. This aspect is important especially with regard to the meshed topology of the European electricity transmission grid.
- The mixed-integer, unit-wise modeling ensures an adequate modeling of the impacts of the unit commitment and dispatch of each German generation unit on resulting physical flows, congestion and redispatch.
- The applied rolling planning approach ensures acceptable computation times and applies the intertemporal modeling of unit commitment for 8760 h without simplifications such as typical hours, leading to a feasible market dispatch (since the impacts of redispatch measures on the availability of generation units are reflected in the dispatch of the following day).

From our point of view and with regard to the focus of our study the simplifications made appear to be an acceptable compromise between the need for a detailed modeling of power plant scheduling and electricity flows on the one hand and manageable calculation times on the other hand.

2.3. Load flow model

As described previously (cf. Section 2.2) in our approach, Germany is represented by 21 aggregated nodes according to 50Hertz Transmission GmbH (2009). Load flow approximation in the scheduling model WILMAR JMM is achieved using PTDFs, together with interface capacities (ICs) derived under N-1 secure operation. In order to calculate PTDFs and ICs for the interfaces between the (aggregated) nodes and to calculate NTCs for the modeling of commercial exchanges (between alternative bidding zones in Germany), a nodal load flow model for Germany is used. In the load flow model, the German extra high voltage transmission grid (380 kV, 220 kV) is represented by 601 nodes (454 regular and 147 auxiliary nodes). All transmission lines and nodes are based on several publicly available grid maps (cf. ENTSO-E 2012a; TU Delft, 2013; Umweltbundesamt, 2013; VDE, 2012). The electrical parameters of the transmission lines are estimated from the lengths, voltage levels, typical impedance values and typical thermal limits for overhead lines (cf. Kiessling et al., 2003). All generators with an installed capacity greater than 100 MW, offshore and onshore wind generation, and the locational vertical load levels are included. The surrounding countries are represented by country nodes that are connected to the German grid via cross-border lines. For a more detailed description of the German load flow model used and of the calculation of PTDFs and NTCs see Bucksteeg et al. (2015).

2.4. WILMAR joint market model

In the following we describe the general characteristics of the WILMAR JMM and therefore focus on the optimization approach applied in both model variants. Yet, due to reasons of clarity, the load flow equations are only described for model variant MIP JMM DE.

The WILMAR JMM, a stochastic scheduling tool used to analyze the impact of the fluctuating feed-in from wind in energy markets, was originally developed in the project Wind Power Integration in Liberalized Electricity Markets (WILMAR) supported by the EU (cf. Barth et al., 2006). The objective function being minimized is the overall variable cost of the system over the optimization period, covering fuel, CO_2, start-up and further variable costs. Multiple technical restrictions e.g. start-up time, minimum up and down times, ramping rates, minimum and maximum generation and reserve provision are included. Besides electricity, heat demand of 8760 h a year also has to be met. The modeled market prices reflect the marginal generation costs. More detailed information about the scheduling model, including all equations and a description of the rolling planning approach, can be found in Tuohy et al. (2009), Weber et al. (2009), Barth et al. (2006) and Meibom et al. (2006).

In this study, perfect foresight is assumed. This means that neither load, wind or solar forecast errors, nor random generation or transmission outages are considered. This is mainly for modeling the pure effects of grid constraints on redispatch. One may also argue that in practice, intraday trading (which is not explicitly taken into account in the model), and reserve markets, handle large parts of power imbalances.

The planning horizon of the WILMAR JMM is up to 36 h with hourly optimization. Therefore, two decision problems can be distinguished (cf. Fig. 2):

1. Day-ahead market for power trading (according to EPEX-based trading)
2. Redispatch to relieve the remaining congestion in the transmission grid (grid-constrained optimization)

As the WILMAR JMM has been developed as a stochastic planning tool with the aim of investigating the impact of fluctuating wind production, a rolling planning concept has been applied in the model. In brief, the power plant dispatch is optimized every 12 h, based on new available information as it arrives. Although the improved accounting for uncertainties (e.g. for wind forecast errors) is not a focus of this paper, the concept of rolling planning is advantageous for investigating the effect of market splitting on redispatch. To put it plainly, it makes a difference whether the hourly (re)dispatch is optimized an entire year in advance or whether it is planned based on a rolling approach. The latter optimizes (re)dispatch every 12 h – based on the last optimized system states. Fig. 2 illustrates the applied two-stage optimization approach with rolling planning of the WILMAR JMM.

The first step represents the day-ahead market. Based on several input parameters the model minimizes the overall costs. Hence, it calculates the cost-optimal day-ahead schedules of the power plants and determines the corresponding trading flows. Then, in the second step (after day-ahead market closure), transmission grid restrictions are taken into account. Thereby the physical flows are calculated using PTDFs. In the event that resulting physical flows exceed the transmission capacities (ICs), this is interpreted as congestion and will be relieved by redispatch, i.e., by the overall cost-minimizing up and down regulation of units compared to the day-ahead schedules. The applied rolling planning approach ensures that the operating states of the units after redispatch optimization are considered in the day-ahead optimization when rolling one time segment (12 h) forward.

The applied model approach corresponds to a cost-based redispatch mechanism because the optimal combination of positive and negative redispatch measures is represented by the least-cost up and down regulation of units in order to relieve grid congestion (cf. Fig. 2, 2nd step), while at the same time meeting the demand

for electricity. Negative redispatch by conventional hydro-thermal units implies cost savings as generation costs are avoided. Positive redispatch induces additional generation costs for a unit's upward generation. It follows that the overall redispatch costs are given by the overall marginal costs of all least-cost plants, which are regulated upwards (positive redispatch) and the difference of the total savings from downward regulated plants (negative redispatch).

In the following, the applied load flow equations are described. These are the main equations that are relevant for the modeling of redispatch in the model variant MIP JMM DE. Descriptions of the indices, parameters and decision variables used can be found in Table A.1 in the Appendix.

To ensure that the trading flow $P^{TRANS,DA}_{r,r',t}$ planned at the day-ahead market (1st optimization step) does not exceed the day-ahead trading capacity $L^{TRANS,DA}_{r,r',t}$ between bidding zones, Eq. (1) is defined:

$$P^{TRANS,DA}_{r,r',t} \leq L^{TRANS,DA}_{r,r',t} \quad \forall\,(r,r') \in V,\ \forall\, t \in T \tag{1}$$

After the day-ahead optimization of the power plant dispatch, physical transmission capacities are taken into account in the second optimization step (cf. Fig. 2). To calculate the physical power exchange between (aggregated) nodes we use the PTDF approach that translates the financial transactions between bidding zones (which result from the day-ahead optimization of the power plant dispatch) into physical load flows. Redispatch measures are needed in the event that physical flows resulting from the day-ahead dispatch exceed the ICs. In the grid-constrained optimization, equations therefore guarantee that the physical flows do not exceed the transmission capacity $L^{TRANS,GC}_{r,r',t}$ for each interconnection. In the second step of the optimization, the trading flows $P^{TRANS,DA}_{r,r',t}$ between (aggregated) nodes (resulting from the day-ahead dispatch) are fixed and only the variables representing positive and negative redispatch (marked by the subscript RD^{+} respectively RD^{-}) are subject to optimization. In the model, DC connections are also considered. This leads to the following key restrictions for the second optimization step:

The net day-ahead trade balance per (aggregated) node r^* inside an AC synchronous area a is given by the sum of the imports and exports on AC transmission lines (resulting from day-ahead optimized trades):

$$P^{TRADE,DA,AC}_{r^*,t} = \sum_{r''|(r^*,r'')\in V^{AC}_a} P^{TRANS,DA,AC}_{r^*,r'',t} - \sum_{r''|(r'',r^*)\in V^{AC}_a} P^{TRANS,DA,AC}_{r'',r^*,t} \quad \forall\, r^* \in R_a,\ \forall\, t \in T \tag{2}$$

The net redispatch balance per (aggregated) node r^* is given by totaling the transmission of power on AC transmission lines resulting from positive and negative redispatch measures (resulting

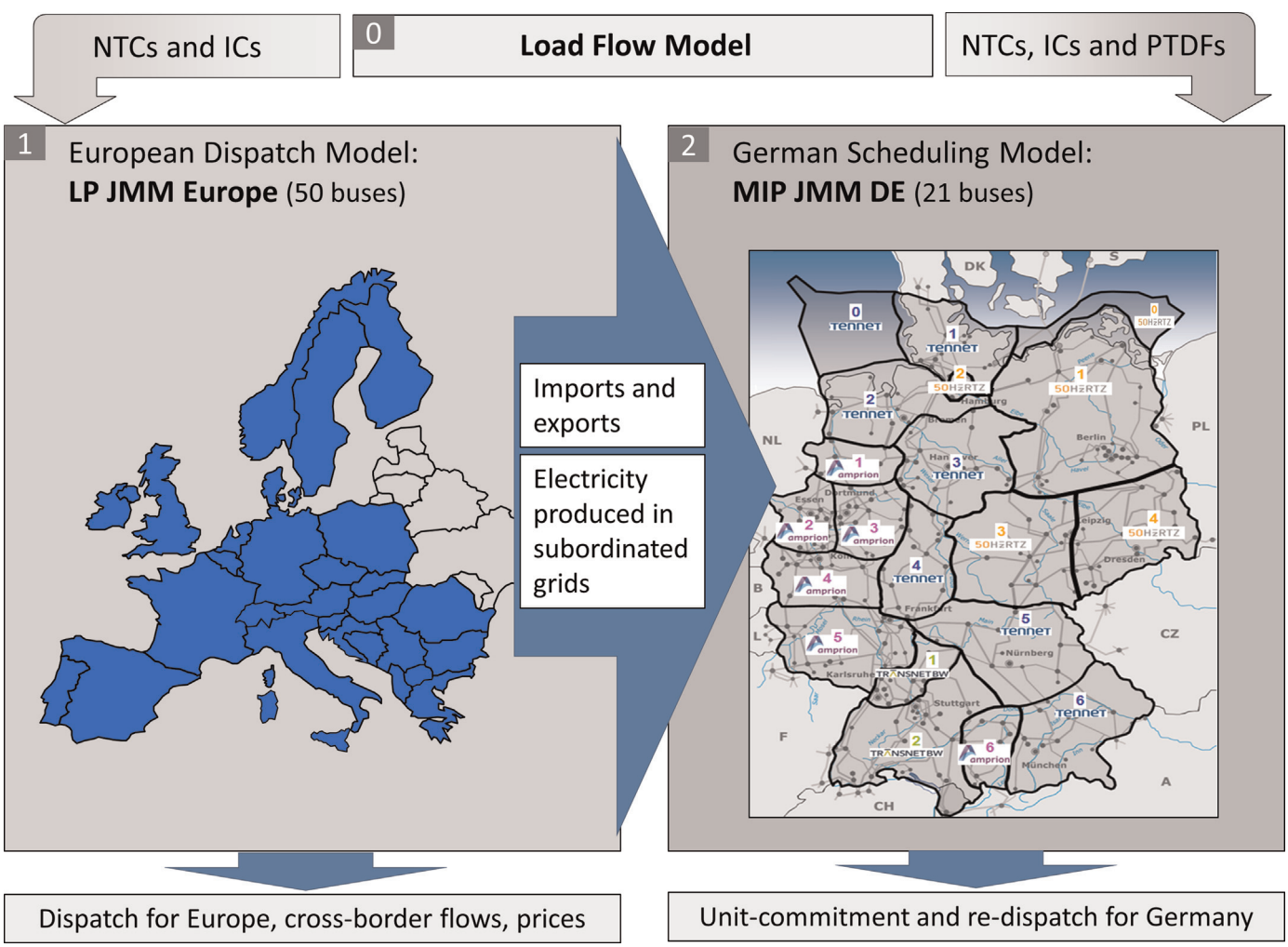

Fig. 1. Basic model framework and geographical scope of the case study.
Note: Approximation of the German transmission grid according to the regional electricity transport model of the German TSOs (cf. 50Hertz Transmission GmbH et al. 2009).

from grid-constrained redispatch optimization):

$$P_{r^*,t}^{NET,GC,AC} = \sum_{r''|(r^*,r'')\in V_a^{AC}} \left(P_{r^*,r'',t}^{TRANS,GC,AC,RD^+} - P_{r^*,r'',t}^{TRANS,GC,AC,RD^-}\right) - \sum_{r''|(r^*,r'')\in V_a^{AC}} \left(P_{r'',r^*,t}^{TRANS,GC,AC,RD^+} - P_{r'',r^*,t}^{TRANS,GC,AC,RD^-}\right) \quad \forall\, r^* \in R_a,\ \forall\, t \in T \tag{3}$$

Using Eqs. (2) and (3) the limiting AC flow constraint can be formulated as follows:

$$\sum_{r^*\in R} PTDF_{r,r',r^*,t} * P_{r^*,t}^{TRADE,DA,AC} + \sum_{r^*\in R} PTDF_{r,r',r^*,t} * P_{r^*,t}^{NET,GC,AC} \le L_{r,r',t}^{TRANS,GC} \quad \forall\, (r, r') \in V_a^{AC},\ \forall\, t \in T \tag{4}$$

As shown by Eq. (4), the AC transmission flows on connection (r, r') resulting from the day-ahead optimized trades and from the redispatch measures (as optimized in the second grid-constrained optimization step) are calculated using the PTDF approach and are limited by the capacity restriction for the connection (r, r').

For DC connections no PTDF calculation is needed and consequently the following Eq. (5) holds in the second optimization step:

$$P_{r,r',t}^{TRADE,DA,DC} + \left(P_{r,r',t}^{TRANS,GC,DC,RD^+} - P_{r,r',t}^{TRANS,GC,DC,RD^-}\right) \le L_{r,r',t}^{TRANS,GC} \quad \forall\, (r, r') \in V^{DC},\ \forall\, t \in T \tag{5}$$

Analogous equations are defined to limit flows in the opposite direction.

2.5. Evaluation criteria for congestion situation and economic impacts

In order to evaluate the potential benefits of market splitting in Germany in 2020, we define both congestion related and economic indicators in the following. We identify four main indicators to measure the impact of market splitting on imminent congestion and corresponding redispatch measures of the German transmission grid for 2020: number of congestion hours, number of congestion event hours, total congestion amount and total redispatch amount. These indicators focus on imminent overloads (congestion) resulting from day-ahead schedules and the adjustments in dispatch (redispatch) needed to resolve these overloads. As the adequate consideration of further grid security related aspects, such as the sufficient provision of reactive power or the maintenance of frequency stability require a detailed AC load flow model, we do not investigate these aspects in our model approach and thus do not consider them in the following.

- *Congestion hours* are those hours in which curative congestion management is needed to preserve system reliability – at least somewhere in the transmission system. The maximum number of congestion hours is 8760, i.e. the number of hours in a year. For example, if there are two congested lines in one hour, only one congestion hour is counted.
- *Congestion event hours* (CEH) are those hours in which redispatch of day-ahead schedules is needed to preserve system reliability. The number of congestion event hours can exceed the number of congestion hours as each congestion event on each transmission line is counted separately. For example, if there are two congested transmission lines in one hour, they are counted as two congestion event hours.
- The *congestion amount* refers to the amount of energy scheduled in the day-ahead market which cannot be transported due to limited physical transmission capacities. Or, stated more plainly, the congestion amount corresponds to the amount of imminent line overloading caused by the day-ahead market schedules. For instance, if the physical transmission capacity at a specific interconnection is limited to 2000 MW, but the day-ahead market schedules lead to a physical flow of 3000 MW across this border, the congestion amount is 1000 MWh in the relevant hour.
- The *redispatch amount* refers to the amount of energy that has to be shifted to remove congestion and preserve system stability. Therefore the amount of downward regulation to relieve a bottleneck corresponds to the amount of upward regulation, as the power system has to be in balance at all times.

It is worthwhile mentioning that the magnitude of adjustments of the day-ahead market schedules (redispatch amount) can be larger than the congestion amount. This is due to Kirchhoff's laws and the spatial distance between available generation units for redispatch and imminent line overloading. For instance, if day-ahead schedules lead to an imminent line overloading of 1000 MW, the congestion might be resolved by the down-regulation by 1000 MW of a generation unit having a high impact on this congested line. However, if this specific unit is not running, a unit with lower impact on the congested line needs to be redispatched, for example by 1500 MW, in order to resolve congestion. As there is no way to achieve a perfect zonal market design with aggregated bidding zones and *ex ante* intra-zonal congestion management, remaining congestion and redispatch are mostly related to internal bottlenecks. Thereby, the availability of generation units for redispatch, and hence the effectiveness of redispatch measures, are affected by the initial day-ahead dispatch, which, in Germany, changes when introducing additional bidding areas. As the amount of positive redispatch corresponds to the amount of negative redispatch, all figures and tables that refer to redispatch volumes in the following apply to negative redispatch unless specified otherwise.

In order to analyze economic effects related to future congestions, we focus on the following economic indicators: total redispatch costs, specific redispatch costs, mean day-ahead price (difference) and total system costs.

- In the described model approach, *total redispatch costs* can be calculated as the difference between total system costs of a simulation with transmission constraints and a simulation without transmission constraints (copperplate). While the copperplate-variant indicates the system costs for a scenario without any binding transmission constraints (in the second optimization step, cf. Fig. 2), the higher system costs in the constrained-variant occur due to redispatch measures needed to remove imminent overloads resulting from day-ahead schedules. It should be noted that the redispatch costs obtained have to be regarded as lower boundary mainly due to cost minimization under perfect foresight and the assumed perfect coordination between the German TSOs.
- *Specific redispatch costs* are then calculated by taking the total redispatch costs divided by the total amount of redispatch (I) or the total amount of congestion (II).
- The *mean day-ahead price difference* is calculated as the price difference from the volume weighted yearly average day-ahead prices of both German bidding zones.
- To derive statements about the effect of market splitting on the *total welfare*, we consider the difference between the *total system costs* with and without market splitting. Costs for the curtailment of wind power generation are not included in the total system costs since these have been set deliberately and arbitrarily high to make wind shedding the ultimate ratio in dispatch

(as requested by the German renewable legislation). Furthermore, effects on *consumer and producer rents* are analyzed.

2.6. Analyzed scenarios and splitting of Germany

We consider two scenarios in this paper: the reference scenario and the market splitting scenario, both for 2020. Load and RES generation profiles for the market simulations are based on the reference year 2012. Furthermore, delays in grid extension of three years are assumed (cf. BNetzA, 2012), which represents an extreme scenario regarding the planned grid development. A detailed description of the input data used can be found in the Appendix, where we also discuss the adequacy of the reference year used and corresponding implications for our findings.

The aim of the reference scenario, as a kind of business as usual scenario, is to analyze and understand challenges arising and opportunities for German TSOs. The focus is therefore on congestion and redispatch associated with the increased RES production and the expected status of network expansion in Germany in 2020. In the reference scenario, it is assumed that the current market design in Germany is maintained until 2020 and congestion management is achieved via cost-based redispatch. While transmission restrictions within Germany are not considered in the day-ahead market (or in other words, Germany is treated as a copperplate), exchange flows between countries are limited via NTCs.

In light of the challenging network situation in Germany, the market splitting scenario investigates the impacts of introducing additional bidding zones as one possible measure to deal with upcoming congestion in the German (and European) electricity system. When defining a market splitting scenario for Germany, two main challenges arise. The first challenge is the specific zonal delimitation. The zonal border (where the day-ahead market is split) should run along the main bottlenecks to incorporate major congestions already in the commercial exchanges on the day-ahead market and thus reduce imminent overloads and redispatch. Otherwise market splitting would not be effective. The second challenge is then to determine the zonal transfer capacity (here: NTC) that is provided for trading on the day-ahead market. In this context we are facing a trade-off between market liquidity and the effectiveness of congestion management. The lower the zonal transfer capacity given to the day-ahead market, the higher the effectiveness of market splitting, as trading flows and resulting physical flows between the bidding zones are more limited. At the same time a transfer capacity between bidding zones that is too low decreases the liquidity and efficiency of the day-ahead market and results in potentially unused transmission capacity along the zonal border.

So far, few studies have addressed the optimal zonal delimitation of the German or the European power market (cf. e.g. Breuer et al. 2013; Burstedde, 2012; Duthaler, 2012; Olmos and Pérez-Arriaga; 2008, Stoft, 1997). The ongoing consultation process on the CACM (which includes the adequacy of bidding zone configuration) at the European level also demonstrates the relevance of this topic (cf. European Commission, 2015). However, this paper does not aim to find the optimal delimitation of alternative bidding zones. It rather aims to present a suitable way to model redispatch and investigate the potential contribution of market splitting to congestion management in Germany. We therefore decided to determine alternative bidding zones for Germany on the basis of the main bottlenecks in the German transmission grid as identified in our model results for 2020. Consequently the scarcity of conventional generation capacities in South Germany is implicitly considered. To model commercial exchanges between the defined bidding zones in the day-ahead market, an NTC for a representative system state is calculated. While Egerer et al. (2015) and Thema Consulting Group (2013) both apply heuristic approaches for the determination of NTCs we implement the methodology described in Bucksteeg et al. (2015). The market splitting scenario is supplemented by two sensitivities in order to assess the impact of the NTC on the market outcomes.

Except for the market splitting, all specific data assumptions made in the market splitting scenario, concerning, for example, the installed generation capacities, power demand and network expansion, are the same as those applied in the reference scenario.

3. Results and discussion

Keeping the focus of this paper in mind, our model results in the following paragraphs are presented and discussed with respect to Germany. However, as the European power system is highly integrated, changes in the German electricity market design also affect price levels and the corresponding cross-border flows within the entire ENTSO-E grid. Hence, the impacts of market splitting in Germany on European countries are also discussed.

3.1. Zonal delimitation and calculation of the net transfer capacity

As mentioned in Section 2.6 we split the German bidding zone along the main bottlenecks in the German transmission grid as identified under the reference scenario for 2020. Thus, Fig. 3 shows the geographical distribution of congestion by interface obtained in this scenario for Germany. The German transmission network is already increasingly stressed, especially in the north–south direction, by the high wind infeed in the northern area. Our model results indicate that new conventional generation capacities and increasing RES infeed in Northern Germany, together with the shutdown of power plants in Southern Germany increasingly result in day-ahead schedules that correspond to load flows exceeding transmission capacity in a north–south direction. During recent years, the most congested line was the 'Thüringer Waldleitung' (interface 50Hertz3/TenneT5, cf. BNetzA and BKartA, 2014). Our model results confirm that this line will still be congested in 2020, as the planned expansion of this line is inadequate. Moreover, new congestions will occur at the interfaces Amprion4/Amprion5 and TenneT5/TransnetBW1 in 2020 (cf. Fig. 3). The greatest level of negative redispatch (i.e. downward regulation of units) is accordingly found within the north of Germany.

In light of these model results and to deal with this increased congestion we decided to split the German electricity market into two bidding zones – a northern and a southern zone (white dashed line in Fig. 3) – along two major bottlenecks 50Hertz3/TenneT5 and Amprion4/Amprion5. As market liquidity generally decreases with more and smaller bidding zones, we decided not to consider a further market splitting along the bottleneck TenneT5/TransnetBW1. It is worth mentioning that our model results show that congestion within this network area is also substantially removed by the described north–south market splitting. The identified delimitation is also in line with previous studies (cf. Consentec and Frontier Economics, 2011; Breuer et al., 2013 and Egerer et al., 2015).

The analysis performed with the load flow model results in an NTC of 12.8 GW for transactions from DE_N to DE_S.[8] In order to explore sensitivities of the market splitting scenario, NTCs of 10.2 GW (−20%) and 15.3 GW (+20%) were considered. As physical flows from DE_S to DE_N are not expected to induce

[8] Thema Consulting Group (2013) analyze two scenarios for 2012 considering a NTC of 7 and 10 GW, respectively. Egerer et al. (2015) apply a NTC of 8 GW for 2015 in their reference case. The incorporation of planned grid expansion leads to an increase of the NTC until 2020.

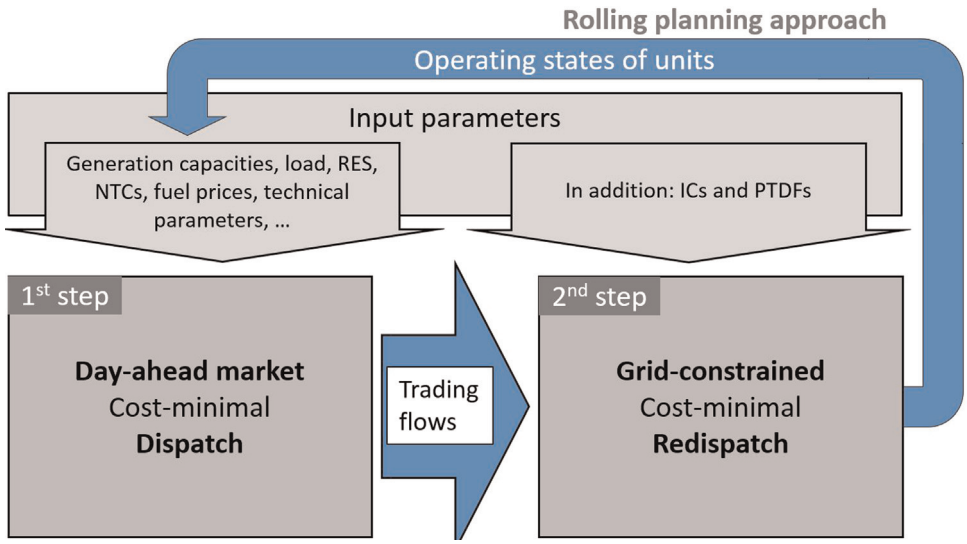

Fig. 2. Two-stage optimization approach (with rolling planning) of the WILMAR JMM for redispatch optimization.

congestion in 2020, commercial exchanges from Southern to Northern Germany were not limited in the day-ahead market.

3.2. Impacts of market splitting on congestion and redispatch

In the previous Section 2.5 we defined several indicators with a focus on congestion and redispatch. Based on the defined indicators, Table 1 summarizes our model results and compares them to real data for 2012.

In general, a shift in bottlenecks can be observed when comparing 2012 to 2020. Besides already existing bottlenecks in 2012 (50Hertz3/TenneT5), new bottlenecks (Amprion4/Amprion5 and TenneT5/TransnetBW1) contribute to substantial increases in congestion and a nearly tripling of the total redispatch volume in 2020 compared to 2012. One of the key reasons is the increase in RES production – and the resulting transport. This includes high cross-border loop flows[9] from the north of Germany through Poland and Czech Republic as well as through the Netherlands, Belgium and France to the south of Germany. Besides additional onshore wind capacities, considerable offshore wind capacities are also included at 2020. The congestion mainly arises from the transport of wind production in the north of Germany to the load centers located in the west and south. Another key reason for congestion (mainly at Amprion4/Amprion5) will be the shutdown of the nuclear plant Philippsburg in 2019, which is located at the regional border Amprion5/TransnetBW1 (cf. Fig. 3). This finding supports the need for the planned High Voltage Direct Current (HVDC) line between Amprion2 and TransnetBW1 'Osterath – Philippsburg' (cf. 50Hertz Transmission GmbH, 2014b). While this line is not included in our model due to the expected delays in grid expansion, it could reduce but not completely relieve the congestion between Amprion4 and Amprion5.

While the number of the modeled congestion event hours (CEH) increases by 40% in the reference scenario 2020 compared to 2012, the modeled redispatch amount increases by 196%. All indicators shown in Table 1 suggest that the network situation in 2020 will intensify the TSOs' challenges to secure the grid operation. In other words, with regard to congestion and redispatch, the power system will tend to be more stressed in 2020 compared to 2012 especially if the planned HVDC lines from north to south are not available by then.

However, since there are only redispatch event hours (REH) publicly available for Germany, the direct comparison of the REH for 2012 with the modeled CEH in 2020 has to be done with care. The main difference is that REH also include congestion events caused by, for example, a sub-optimal activation of control energy. This might be induced by the fact that there is no integrated balancing and redispatch market in Germany in place.

The model results for both scenarios in 2020 clearly indicate that market splitting in Germany contributes to a more secure system operation since it lowers the total congestion amount and the need for (manual) curative congestion management. While the total number of congestion hours in 2020 can be reduced by 3084 h (or 56%) through market splitting, the number of CEH decreases by 68% (compared to the reference scenario 2020). The corresponding congestion amount shrinks by 83% and the total redispatch volume decreases by 69%.

Fig. 4 shows where congestion in Germany occurs in both scenarios. As expected, the major reduction in terms of congestion occurs along the zonal border. It is noticeable that the relative reduction of intra-zonal congestion is comparable to that of inter-zonal congestion. One could have expected that the relative reduction would be higher at zonal borders, yet there are indirect effects. For example, if congestion caused by high wind feed-in in the north is removed by market splitting at the zonal border DE_N/DE_S, this will also reduce the transmission flows within network areas situated further south. Consequently congestion within these areas is lower.

The corresponding (negative) redispatch amount that is needed to remove congestion can be reduced by 5 TWh (69%) through market splitting (cf. Fig. 5). The reduction is mainly achieved within the northern bidding zone DE_N since congestion generally occurs in a north–south direction.

While market splitting primarily deals with inter-zonal congestion, redispatch is still required within the two bidding zones to correct the day-ahead market schedules and to remove intra-zonal congestion. This can be explained by intra-zonal physical transmission constraints that are enforced in the second optimization step and are not considered in the day-ahead market (cf. Fig. 2). That is why the impact of a further reduced NTC on remaining redispatch requirements under the "NTC −20%" sensitivity is limited. As the curtailment of wind generation in the North of Germany can be attributed almost solely to intra-zonal transmission constraints, market splitting does not prevent the curtailment of wind generation and thus does not improve the integration of wind energy in this regard.

Due to the unbalanced use of transmission lines along the zonal border, some congestion also remains between the day-ahead

[9] For a distinction of loop and transit flows see Schavemaker and Beune, 2013.

Table 1
Comparison of congestion and redispatch indicators for Germany for 2012 and 2020.

Indicator	Unit	Historical Germany 2012	Modeled Germany 2012	Reference scenario, Germany 2020	Market splitting scenario, Germany 2020
Number of congestion hours	Hours	n.a.	4835	5535	2454 (−56%)
Number of congestion event hours (CEH)	Hours	7160[a]	6324	8910	2859 (−68%)
Total congestion amount	GWh	n.a.	2324	7053	1165 (−83%)
Total redispatch amount	GWh	2566	2499	7392	2301 (−69%)

Sources: BNetzA and BKartA (2014), TSO websites, own calculations

[a] Number of hours with electricity flow related redispatch measures as per §13.1 EnWG (German Energy Law).

bidding zones (about 4% compared to the reference scenario 2020, cf. Fig. 4). Under the "NTC −20%" sensitivity, inter-zonal congestion is almost fully removed. Hence, the amount of inter-zonal congestion captured on a day-ahead basis depends largely on the considered cross-zonal transfer capacity. The lower the zonal transfer capacity, the smaller the day-ahead trade in energy between the bidding zones and the lower the probability that resulting exchange flows exceed the physical transmission capacities. The appropriate choice of the zonal transfer capacity hence has a strong influence on the effectiveness of market splitting (cf. sensitivities NTC +/−20%). Only an appropriate design of the market splitting can contribute to system security. However, there will still be the need for some redispatch of the system as transmission constraints may be violated by intra-zonal power flows (cf. Hogan, 2012).

Regarding cross-border effects of market splitting, the model results indicate that adequate price zones would reduce unplanned transit and loop flows via interconnected countries. This can be explained by a more even geographical distribution of electricity generation through market splitting, as the generation of coal plants with low marginal costs in DE_N and the Netherlands is substituted by the generation of coal and gas plants with higher marginal costs in DE_S, France and the Czech Republic. However, this might not hold in general and largely depends on the design of the market splitting.

3.3. Economic impacts of market splitting

In this section we discuss the economic effects of market splitting based on the defined economic indicators in Section 2.5. Table 2 summarizes the model results for 2020 and compares them to real data for 2012 (as far as data is publicly available).

The increase observed in the modeled redispatch volumes from 2012 to 2020 implies higher total redispatch costs (+38%). Interestingly, the specific redispatch costs I (per $MWh_{redispatch}$) are lower in 2020 than 2012. This can be explained principally by the disproportional increase of redispatch volumes. Moreover, historical redispatch costs might be driven by an imperfect coordination between the German TSOs and by inefficiencies caused by a non-optimal redispatch, for example due to forecast errors. In contrast, we assume complete information in our model and thus a perfect coordination between TSOs and an optimal redispatch.

The increase over 2012 in the mean day-ahead price of about 4 €/MWh obtained in the reference scenario 2020 is mainly a result of a higher CO_2 price. Consequently spot prices increase in hours with low residual electricity demand, as conventional power plants (especially lignite-fired) with higher marginal costs are setting the market price. This effect is strengthened by the shutdown of conventional (especially nuclear) power plants by 2020.

In line with the reduction of congestion and redispatch volumes, market splitting induces a substantial decrease in total redispatch costs of 64% compared to the reference scenario 2020. However, specific redispatch costs increase (cf. Table 2). This can be explained by a reduced effectiveness of redispatch as a consequence of differing day-ahead schedules in the market splitting scenario. Thus, the remaining intra-zonal congestion in DE_S is relieved by a down regulation of lignite plants rather than hard coal plants located in DE_N. The higher spatial distance between redispatched lignite plants and congestion results in a lower effectiveness in reducing load flows on the congested lines. As a consequence, specific redispatch costs II are twice as high as specific redispatch costs I.

Market splitting only affects prices in those hours in which congestion occurs and the NTC of 12.8 GW becomes binding. This is the case in 3109 h in 2020 and leads to a difference of 2.39 €/MWh in annual average prices between both bidding zones (cf. Table 2). The hourly price differences are in the range of 0.05–41.51 €/MWh. This price difference can be interpreted as congestion costs per MWh ignoring transmission losses. Depending on the NTC and, given the +/−20% sensitivities, annual average price differences for 2020 are in the range of 1.55–3.56 €/MWh (cf. Fig. 6). Assuming an NTC of 8 GW Egerer et al. (2015) find an annual average price differential of 1.70 €/MWh for 2015. Thema Consulting Group (2013) compute annual average price differences of 1.50 €/MWh and 3.80 €/MWh for 2012 and show that the price difference between the bidding zones is sensitive to the assumed NTC of 10 and 7 GW, respectively.

Overall, the introduction of market splitting in Germany leads to a negligible reduction of total system costs of only 57 M€ per year in Europe.[10] This is probably an underestimation of the actual cost savings since a cost-efficient redispatch under perfect foresight and perfect coordination between the German TSOs is assumed. Similarly, Kunz (2013) found rather low effects on overall costs for 2020 when comparing uniform pricing with a nodal pricing regime in Germany. However, he also argues that even if the overall costs are equivalent, their distribution across consumers, producers and network operators may change.

Market splitting induces an increase of day-ahead prices in the south of Germany. This is a consequence of limited import transmission capacity (NTC) from DE_N to DE_S, so that wind generation with low marginal cost in DE_N has to be replaced by higher-cost units located in DE_S in order to meet the local demand. At the same time, day-ahead prices in the north of Germany decrease due to the limited export capacities to the south. According to the described price effect, the consumers in DE_N would benefit from market splitting while the producer rents decrease. Conversely, the producers in DE_S would benefit from increasing prices induced by market splitting while the consumer rents would

[10] Costs for wind curtailment are not considered.

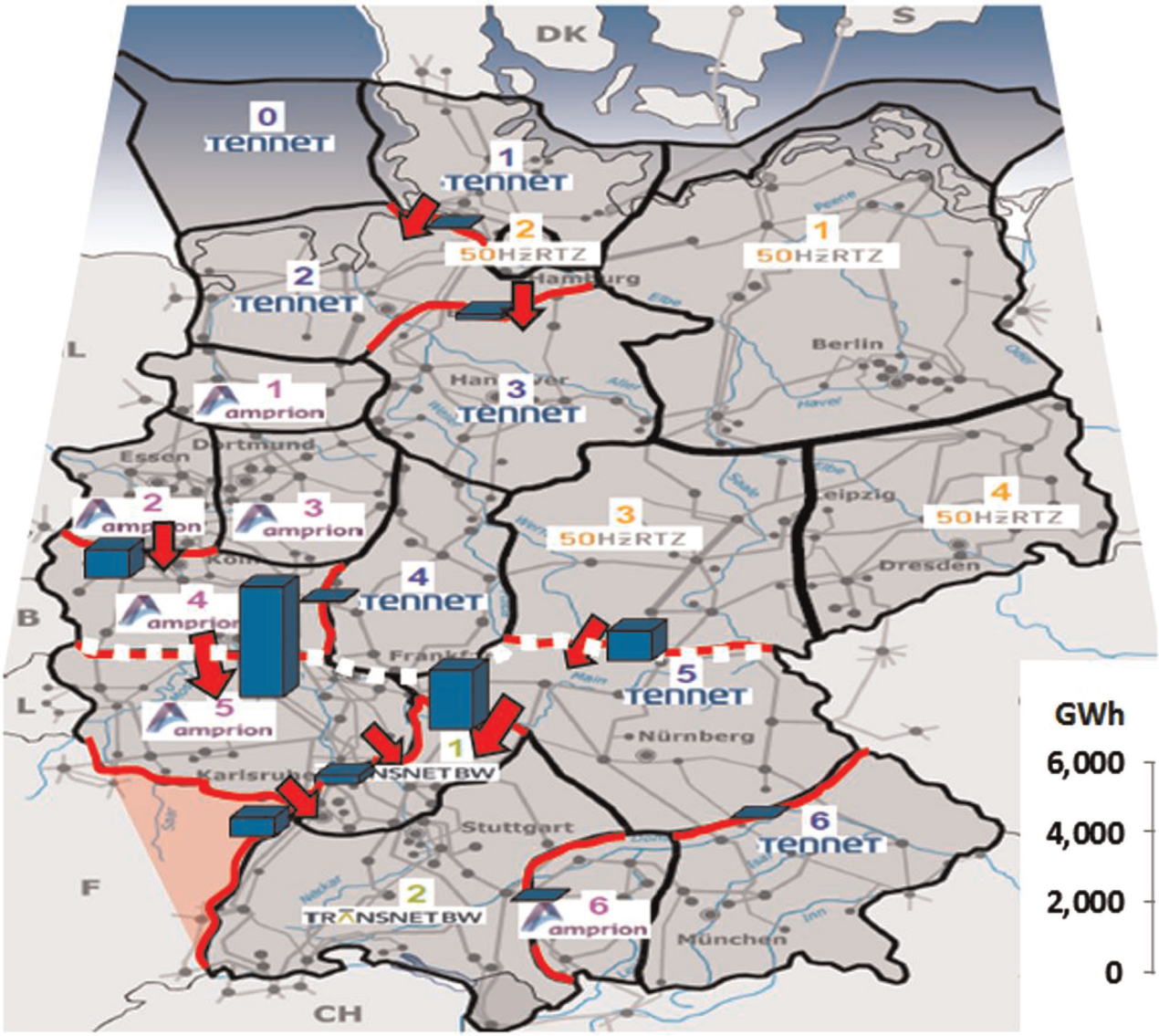

Fig. 3. Regional distribution of congestion (reference scenario 2020).

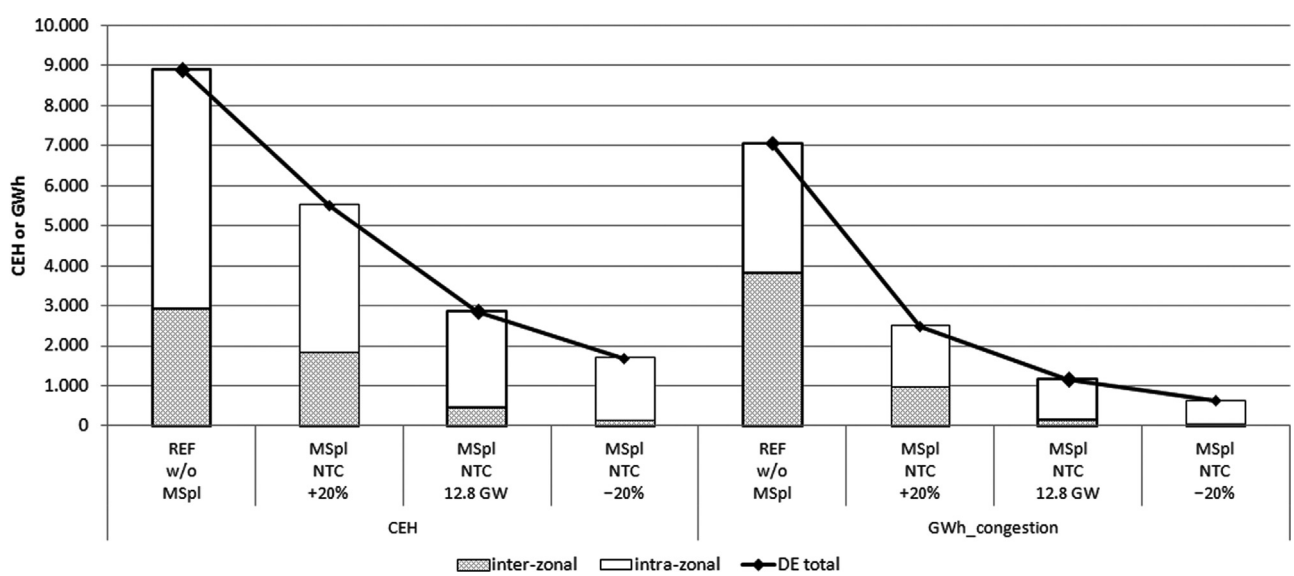

Fig. 4. Impact of market splitting on congestion.

Table 2
Comparison of economic indicators for Germany for 2012 and 2020.

Economic indicator	Unit	Historical DE 2012	Modeled DE 2012	REF DE 2020	MSpl DE 2020	
Total redispatch costs	M€	165[a]	137	190	69 (−64%)	
Specific redispatch costs I[b]	€/MWh$_{redispatch}$	64.24	54.82	25.64	29.90 (+17%)	
Specific redispatch costs II[c]	€/MWh$_{congestion}$	n.a.	58.95	26.87	59.10 (+120%)	
					DE_N	DE_S
Mean day-ahead price DE	€/MWh	42.63[d]	43.47	46.24	46.22	48.62
Change of producer rents	M€	n.a.	n.a.	–	−483	+164
consumer rents	M€	n.a.	n.a.	–	+318	−279
congestion rents	M€	n.a.	n.a.	–	+233	
Total system costs Europe	M€	n.a.	n.a.	73,382	73,325	

Sources: BNetzA and BKartA (2014), www.eex.com, own calculations

[a] includes only costs for redispatch and countertrading.
[b] total redispatch costs divided by total amount of redispatch (cf. Table 1).
[c] total redispatch costs divided by total amount of congestion (cf. Table 1).
[d] yearly average of volume-weighted EEX spot prices.

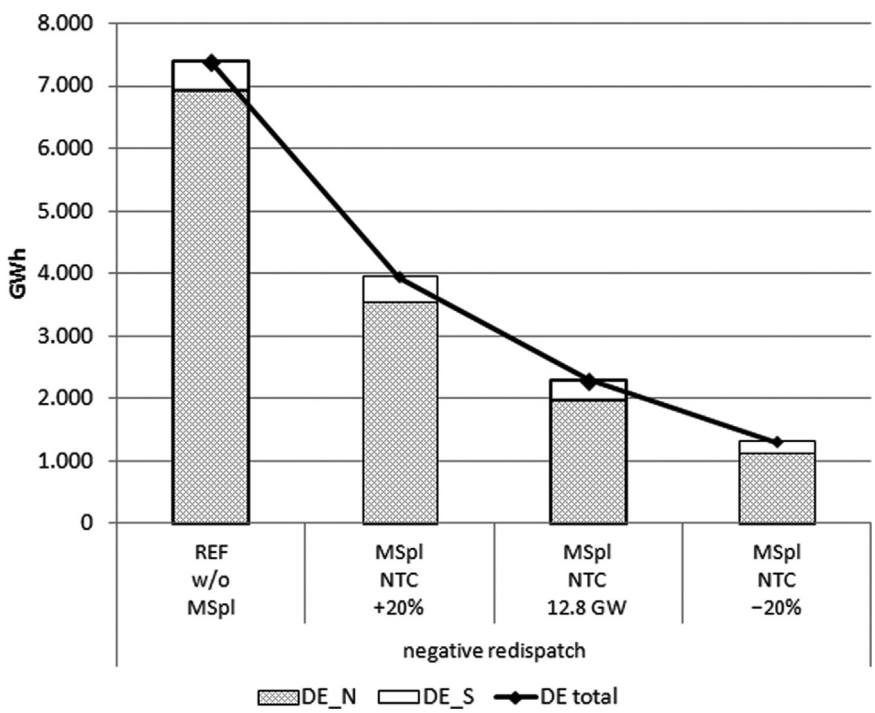

Fig. 5. Impact of market splitting on (negative) redispatch.

decrease (cf. Table 2). Besides wind plants, base load generation technologies (nuclear, lignite and coal) particularly suffer losses of 483 M€ per year in the north region. On the other hand, profits increase by 164 M€ per year in southern Germany, i.e. not to the same extent. The pricing of transmission capacity between DE_N and DE_S in the day-ahead market provides scarcity rents to the TSOs of 233 M€ per year in 2020 (cf. Table 2).

The results under the "NTC +/−20%" sensitivities demonstrate the impact of the applied NTC both on price effects and distributional effects (cf. Fig. 7). The lower the NTC, the higher the price difference and corresponding distributional effects. Therefore, it is noticeable that the relative decrease of producer rents in Northern Germany is higher than the increase of producer rents in the southern zone.

In order to avoid politically undesirable distributional impacts on consumers, *ex post* aggregations of zonal prices for demand could be introduced (i.e. as implemented in Italy). However, load-weighted average prices would counteract the development of demand response in Germany (cf. Sioshansi, 2008).

Regarding the integration of RES in Germany, the described price effects would reduce producer rents and thus the market value of wind generation in Northern Germany where about 80% of the total wind generation capacity is installed. In this regard market splitting would extend and increase the support requirements for wind energy. Conversely, the higher prices in DE_S could intensify investments in advanced wind turbines for weak wind conditions and thus lead to a geographically more balanced distribution of wind generation capacities in Germany. This assumes the current renewable support mechanisms are adjusted accordingly and RES are fully integrated into the power market (cf. Hirth and Mueller, 2014). Consequently, the allocation of the use of scarce transmission capacity would set economic incentives to decrease the spatial imbalance of supply and load.

Due to the coupling of national power systems in Europe, the introduction of market splitting in Germany also affects interconnected power markets. Fig. 8 depicts the resulting price differences between the reference and the market splitting scenario.

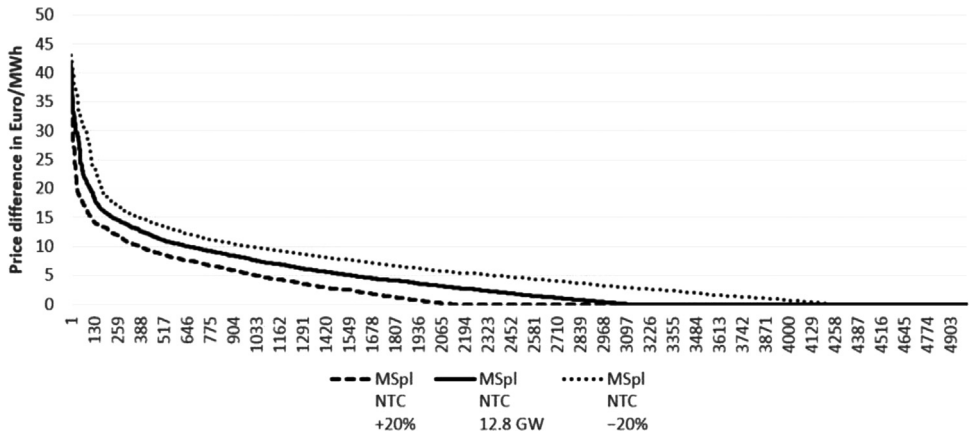

Fig. 6. Impact of market splitting on mean day-ahead prices in Germany.

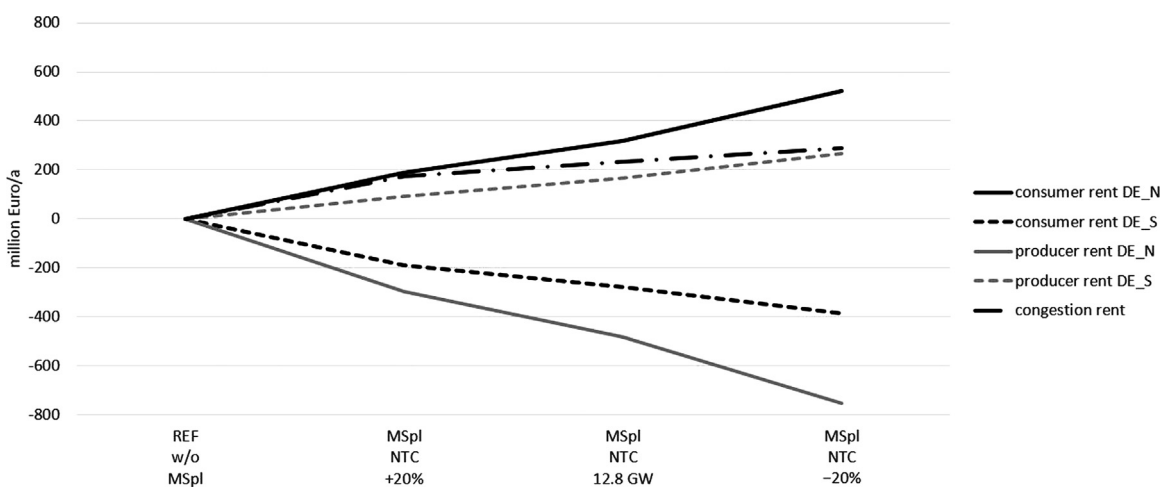

Fig. 7. Impact of market splitting on rents.

While prices in the Nordic countries decrease, increasing day-ahead prices are observed in most countries in continental Europe. As the market splitting leads to low marginal costs in DE_N during periods with high wind infeed, the excess production of wind energy is exported to the Nordic countries. Transit flows from DE_N to the southern countries decrease as another consequence of market splitting, which in turn lead to higher marginal costs in countries that are connected to DE_S.

To summarize, the main benefits of market splitting for Germany are seen in a significant decrease of congestion and redispatch volumes which in turn lowers the risk of contingencies (cf. Section 3.2) and hence contributes to a more efficient system operation. In addition one may argue that, compared to the current German market design with uniform pricing, market splitting could set locational incentives for investments in generation and allow the operation of power plants, especially in Southern Germany where day-ahead prices would increase, to continue.[11] While the overall welfare effect is negligible, price differences between the bidding areas result in considerable distributional effects, which induce political challenges for the affected stakeholders.

[11] A detailed analysis of investment incentives through market splitting is not the primary focus of this study and should therefore be addressed in future research. Harvey and Hogan (2000) already discuss several implications of zonal pricing with regard to the setting of incentives for transmission investments.

4. Conclusions and policy implications

Our analysis indicates that there will be a much more challenging network situation in Germany in 2020. In general, a shift in bottlenecks can be observed. In the case of a single bidding area, new bottlenecks (Amprion4/Amprion5 and TenneT5/TransnetBW1), as well as already existing ones (Thüringer Waldleitung 50Hertz3/TenneT5), lead to a tripling of redispatch volumes in 2020 compared to 2012. The main reasons for this are the expansion of wind and conventional hydro-thermal capacities in the North of Germany, the shutdown of generation capacities in Southern Germany and delays in transmission grid expansion. Together with the high increase in congestion and redispatch volumes, the total redispatch costs increase by 38% by 2020. Besides investments in the transmission network one way of addressing increasing internal congestion is the allocation of the use of scarce transmission capacity by market splitting.

In this regard our model results indicate that an implementation of market splitting with two bidding areas in Germany would generate a significant reduction of total congestion and redispatch volumes. However, due to an unbalanced use of transmission lines along the zonal border some congestion remains in the model. Compared to the reference scenario total redispatch costs are reduced by 64% by 2020. The limited transmission capacity between the northern and the southern German bidding zone results in lower prices in the north and higher prices in the south on

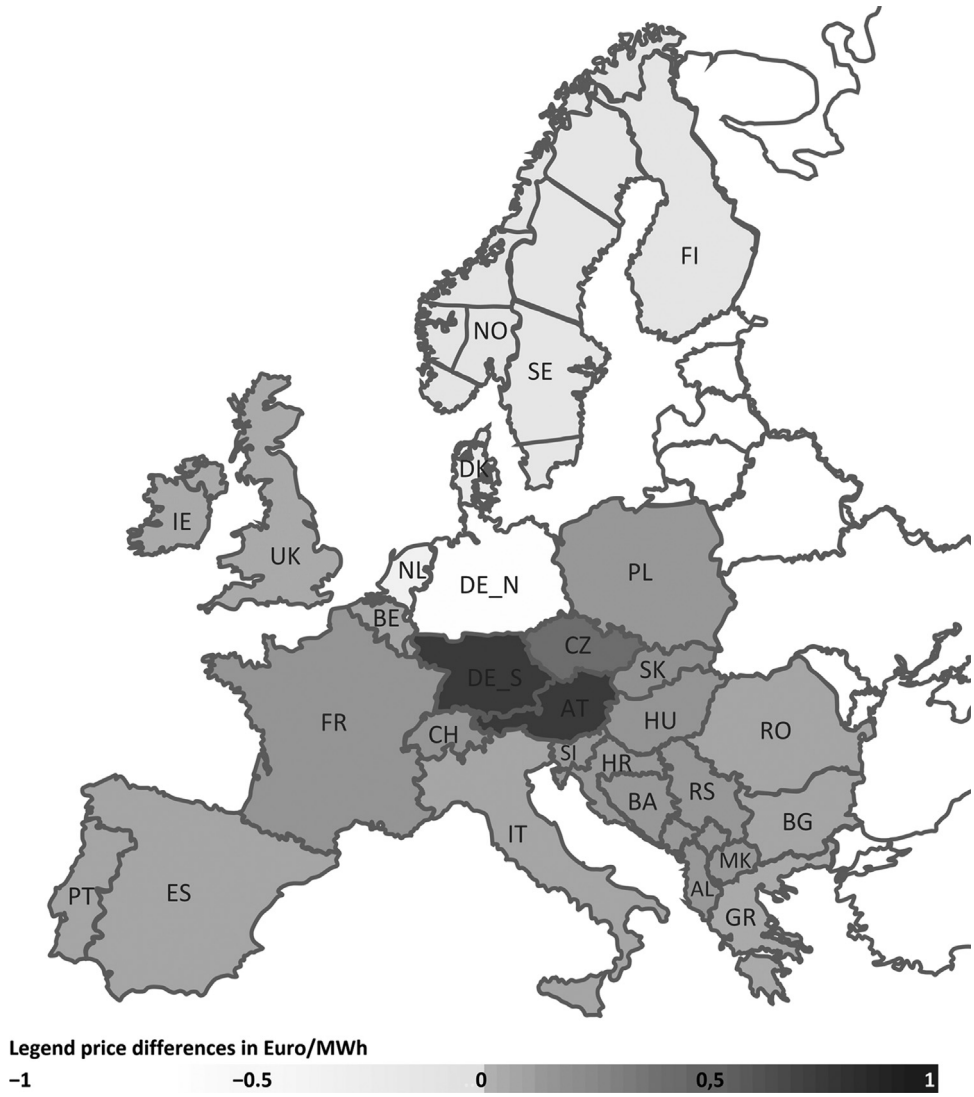

Fig. 8. Impact of market splitting on mean day-ahead prices in Europe.

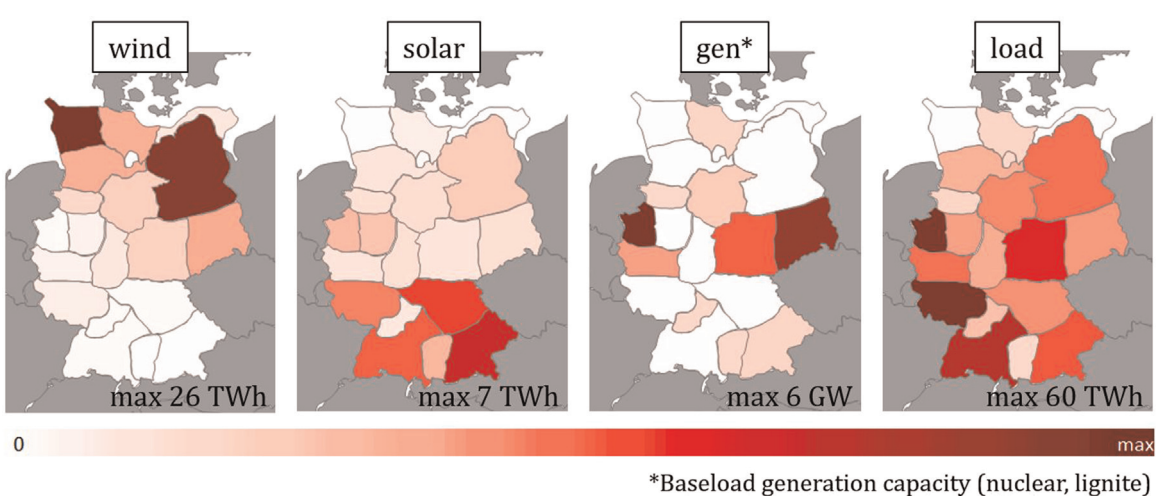

Fig. 9. Regional distribution.

average. The overall welfare effect induced by market splitting in Germany, calculated as a difference in the total system costs, is negligible (decrease of total European system costs by 57 M€). However, the price effects induce considerable distributional effects; consumer rents in the north of Germany and producer rents in the South increase by several hundred million Euros per year and vice versa. In order to avoid politically undesirable distributional impacts on consumers, *ex post* aggregations of locational prices for demand might however be introduced.

The beneficial effects of market splitting depend substantially on its design. This particularly includes (1) the appropriate design of the bidding zones and (2) the adequate determination of the zonal transfer capacity. The basis of the zonal borders should be the main bottlenecks, so that the physical grid constraints are at least partly considered by the zonal transfer capacity in the day-ahead market. Grid investments and entries of new generation units, and thus changing dispatch and congestion patterns, especially require regular checks and adaptations of the bidding zones. Concerning the determination of the zonal transfer capacity, a trade-off between market liquidity and the effectiveness of congestion management arises. If the transfer capacity is set too high, the effectiveness of market splitting would be compromised as day-ahead market schedules still result in line overloading. If the zonal trading capacity is in turn set too low, the actually available physical transmission capacity would frequently not be fully used. As a consequence, trading between the bidding zones and thus market efficiency would decrease. However, there would still be the need for redispatch measures as transmission constraints may be violated by intra-zonal power flows. In this regard, there is no way to a perfect zonal market design as it is not possible to determine aggregated bidding zones and consider intra-zonal impacts on the transmission grid *ex ante* (cf. Hogan, 2012). Only locational marginal prices fully reflect all constraints of the transmission system. Following Harvey and Hogan (2000), nodal pricing is moreover preferred for efficiency reasons and to mitigate market power. Yet, the implementation of locational marginal prices is linked to several institutional challenges (cf. Neuhoff et al., 2013).

In the light of the discussion surrounding the adequacy of the current bidding zone configuration of the European power market, our results also provide a starting point for a possible zonal delimitation, supporting the management of congestion within Germany. Moreover, we define technical and economic measures for the assessment of bidding zone configurations in line with selected evaluation criteria as suggested in the Network Code on Capacity Allocation and Congestion Management (cf. ENTSO-E, 2012b and European Commission, 2015).

To avoid inconsistencies, bidding zones also have to be taken into account in forward and balancing markets. In the case of forward markets, financial transmission rights are one way of assisting market participants in hedging their risks. As the balancing markets are normally monopsonistic, where the system operator is the demander, offers with locational information about the provider of balancing power would help to prevent a sub-optimal congestion intensifying activation of balancing power.

In the long-term, the price effects would improve the integration of RES in Southern Germany, so that a more balanced geographical distribution in Germany, particularly of wind generation capacities, could be achieved. To ensure this, the currently implemented support schemes with feed-in tariffs should be amended and RES should be fully integrated into the power market. Yet, our model results show that market splitting does not prevent the shedding of wind generation in Northern Germany. In the long-term, this effect may be attenuated as higher investments in advanced wind turbines for weak wind conditions (incentivized by higher prices in DE_S) or an increase of internal transmission capacities or interconnection capacities to Scandinavia might improve the integration of wind energy.

In future the increasing share of RES leads to additional flexibility requirements. While flexibility, both at the generation and the demand side is needed to balance intermittent RES infeed, flexibility could also be used to address network issues. Thereby, the introduction of market splitting could enhance the integration of demand side flexibilities if retail prices fully reflect zonal prices. Hence, pricing scarce transmission capacity in the day-ahead market would incentivize consumers to increase their demand in bidding areas with excess supply and low electricity prices (e.g. DE_N) and decrease their demand in market zones with scarce generation and import capacity (e.g. DE_S). Besides market splitting as a market based solution, the pricing of scarce transmission capacity could be used to advance the network tariff design and internalize congestion costs. However, both solutions would require a comprehensive introduction of smart meters and a pooling of flexibilities to become fully effective.

The latter point is also relevant in view of an increasing share of distributed generation from small-scale plants. Large-scale plants connected to the transmission grid are so far mostly used for curative congestion management in Germany. The integration of distributed generation into redispatching would however require an established set of rules for pooling models. The possible level of aggregation particularly depends on the spatial resolution of the zonal (or nodal) market. While flexible load entities and virtual power plants already take part in the balancing markets, bids supplemented by locational information could support the integration of generation and demand side flexibilities into the redispatch system. For long-term development, an integration of balancing markets and congestion management following the example of the Scandinavian Regulating Power Market might be an attractive solution which should be addressed by further research.

To conclude, the results of this paper indicate that additional bidding zones in Germany contribute to congestion management and constitute a solution to reduce redispatch measures as long as transmission grid expansion is further delayed. Beyond that, periodical reviews of the bidding zones as foreseen by the Network Code on Capacity Allocation and Congestion Management are necessary to guarantee an effective and efficient congestion management. Moreover, the economic impacts and related market design issues discussed, as well as potential market imperfections, need to be considered when aiming for an adequate and exhaustive valuation of (alternative) bidding zone configurations. Besides a comprehensive specification of applicable evaluation criteria in line with the Network Code on Capacity Allocation and Congestion Management, further research should address potential market failures, such as market power abuse, and analyze different resolutions of zonal and nodal pricing systems in Europe in this regard.

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Appendix

Detailed data description

The input data used for the case study is described in the following in more detail. There are some parameters which are constant in all scenarios and which are taken from the reference year 2012. These are, in particular, wind and solar profiles, electricity and heat demand profiles, electricity and heat demand levels, water reservoir levels, water inflows and plant availabilities.

Electricity demand:

The reference electricity demand for each country is based on 2012 (cf. International Energy Agency (IEA), 2014 and ENTSO-E, 2012c). Hourly load profiles are taken from ENTSO-E (2012c). For Germany the geographical distribution of the total demand (535 TWh for 2012 and 2020) has been calculated in relation to the distribution of inhabitants by ZIP code and considering large-scale electricity consumption in industrial conurbations. For the regional distribution of the annual total load see Fig. 9. For reasons of simplicity and due to the low economic potential up until 2020, demand side management is not considered in this paper.

As the model results are, among other things, driven by the chosen reference year, we analyzed the adequateness of load profiles from 2006 to 2014. We find a high standard deviation of hourly load values for 2012, which means that high load values are comparatively higher, while low load values are comparatively lower. This finding is particularly driven by seasonal effects with a slightly higher demand during winter (especially February) and lower demand during summer (especially June) in 2012. In general, in the German power system, a high demand induces congestion due to the spatial imbalance of supply and load. However, the described seasonal effects in 2012 have compensatory impacts on the annual congestion amount.

Renewables: wind and solar production

The current German market design related to renewable generation is retained in all scenarios. In particular, this means that the (unlimited) priority of RES feed-in is retained and wind curtailment is only allowed for purposes of system security. Because of the large-scale expansion of RES, it is assumed that the curtailment of wind is also possible on the day-ahead market. With regard to the priority of RES feed-in in Germany, we assume high costs for wind curtailment (1000 €/MW h) to ensure that wind is only curtailed as 'last order' measure in the model (to maintain system security – but at high costs). In general, the development of RES in all European countries is based on the national renewable action plans, for Germany the expectations of 50Hertz Transmission GmbH (2014b) are retained (cf. Table A.2). Wind and solar production are modeled as exogenous production and are specific for each (aggregated) node.

The geographical distribution of solar and wind production for 2020 in Germany is determined proportionally to the distribution of the installed capacities in 2012 (cf. Fig. 9). The distribution of offshore wind production is proportional to the installed capacities of the planned projects: 88% North Sea and 12% Baltic Sea. It has to be noted that the spatial distribution of load and generation is one of the most critical assumptions and a major cause of the congestion situation in Germany.

Due to the incompleteness of publicly available RES data the chosen reference year is based on an analysis of the historical years 2011–2014. Compared to 2011, 2013 and 2014 hourly values of wind generation in 2012 have a lower standard deviation and vary less between the seasons. As new wind turbines reach greater heights, where wind blows more steadily, the lower volatility of wind profiles from 2012 seems to be appropriate for future years (2020). Moreover, we calculate that wind generation is lower during summer 2012 than in the other years. In combination with a higher photovoltaic solar generation in Southern Germany and lower demand during summer this leads to a decreasing impact on congestion.

Compared to 2011, 2013 and 2014 hourly values of photovoltaic solar generation have a lower standard deviation in 2012. During summer (especially June and July) we find a comparatively low photovoltaic infeed, which has an increasing impact on

Table A.1
Nomenclature (JMM).

Indices and sets	Description
AC	Index referring to alternating current line
DA	Index referring to the 1st step of optimization: cost-minimizing optimization of power plant dispatch (day-ahead)
DC	Index referring to direct current line
GC	Index referring to the 2nd step of optimization: grid-constrained cost-minimizing optimization of redispatch
$r,\ r',\ r'',\ r^*$	Index of (aggregated) nodes
R	Set of (aggregated) nodes
R_a	Set of (aggregated) nodes within an AC synchronous area a
t,T	Index and set of time steps
$V,\ V_a^{AC},\ V^{DC}$	Set of (AC and DC) connection lines between (aggregated) nodes
Parameters and decision variables	**Description**
$L_{r,r',t}^{TRANS,DA}$	Maximum trading capacity between r and r' on the day-ahead market at time step t (NTC)
$L_{r,r',t}^{TRANS,GC}$	Maximum transmission capacity of tie lines from r to r' at time step t (IC)
$PTDF_{r,r',r^*,r^{**},t}$	Power transfer distribution factor of the tie lines (r, r') for the real power transaction between r^* (source) and r^{**} (sink) at time step t. For a fixed sink it can be written $ptdf_{r,r',r^*,t}$
$P_{r,r',t}^{TRANS,DA}$	Planned trading flow (on all tie lines) from r to r' resulting from the day-ahead optimized dispatch at time step t (subject to cost-minimizing dispatch optimization)
$P_{r,r',t}^{TRANS,DA,AC}$	Planned trading flows on AC transmission lines from r to r' resulting from the day-ahead optimized dispatch at time step t (subject to cost-minimizing dispatch optimization)
$P_{r,r',t}^{TRANS,DA,DC}$	Planned trading flows on DC transmission lines from r to r' resulting from the day-ahead optimized dispatch at time step t (subject to cost-minimizing dispatch optimization)
$P_{r^*,t}^{TRADE,DA,AC}$	Net day-ahead trade balance per (aggregated) node r^*(inside an AC synchronous area a) given by the sum of the imports and exports on AC transmission lines at time step t (subject to cost-minimizing dispatch optimization)
$P_{r,r',t}^{TRANS,GC,AC,RD^+}$, $P_{r,r',t}^{TRANS,GC,AC,RD^-}$	Transmission of power on AC transmission lines from r to r' at time step t resulting from positive (RD^+) respectively negative (RD^-) redispatch measures (subject to grid-constrained redispatch optimization)
$P_{r,r',t}^{TRANS,GC,DC,RD^+}$, $P_{r,r',t}^{TRANS,GC,DC,RD^-}$	Transmission of power on DC transmission lines from r to r' at time step t resulting from positive (RD^+) respectively negative (RD^-) redispatch measures (subject to grid-constrained redispatch optimization)
$P_{r^*,t}^{NET,GC,AC}$	Net redispatch balance per (aggregated) node r^*given by the sum of the power on AC transmission lines at time step t resulting from positive and negative redispatch measures (subject to grid-constrained redispatch optimization)

Table A.2
Annual RES production 2012 and 2020 in Germany TWh.

RES	2012	2020
Onshore wind	45.2	94.6
Offshore wind	0.7	30.3
Photovoltaic	27.8	51.6
Total	73.7	176.5

Source: 50Hertz Transmission GmbH et al. (2014a).

Table A.3
Installed generation capacities for Germany as considered in MIP JMM DE.

Technology	Installed capacity MIP JMM DE [GW]		Technology	Installed capacity MIP JMM DE [GW]	
	2012	2020		2012	2020
Nuclear	12.2	8.2	Run-of-river	0.05	0.05
Lignite	19.4	16.9	Annual storage	1.1	1.1
Coal	19.5	22.3			
Gas	12.1	13.7	Pumping storage	7.0	9.8
Oil	1.5	0.3			
Biomass	0.2	0.6	Total	73.1	73.0

Note: Only power plants directly connected to the high-voltage-transmission grid or with high relevance for redispatch.

Table A.4
General fuel and CO_2 price assumptions 2012 and 2020.

	Fuel [€/MW h]						CO_2 [€/t CO_2]
	Coal	Lignite	Natural Gas	Nuclear	Fuel oil	Light oil	
2012	10.5	3.9	25.2	2.6	48.8	53.1	7.4
2020	9.9	3.9	24.8	2.6	46.6	72.9	25.0

congestion, as a higher share of photovoltaic capacity is located in Southern Germany with scarce conventional generation capacity.

Conventional power plants:

Commissioning and decommissioning of conventional power plants in Europe are taken into account based on the EWL[12] database, which is continuously updated from publications and press reports. For Germany, the power plant fleet 2020 is considered especially with regard to the power plant list published by the BNetzA (2014). While the total installed capacity is considered in the European dispatch model (LP JMM Europe), the vertical load retained in the German scheduling model (MIP JMM DE) has only to be met by those power plants which are directly connected to the high-voltage transmission grid or which are relevant for redispatch. Table A.3 shows the considered energy mix as input for the German scheduling model MIP JMM DE.

Transmission network:

Given the currently known delays in grid expansion, the timeline for completion of all projects proposed under the national grid development studies and the EnLAG (cf. BNetzA, 2012, 50Hertz Transmission GmbH, 2014a) has been reviewed carefully for Germany. NTC values determining the day-ahead trading capabilities between European countries are taken from ENTSO-E (2014b). As already described in Section 2, PTDFs and IC values (used for load flow approximation within Germany) are derived from the nodal load flow model as described in Section 2.3.

Fuel and CO_2 prices:

Fuel prices (cf. Table A.4) are represented as sum of a general fuel price (wherever possible derived from historical and future prices) and a country-specific transportation cost component.

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